Bank Loan Analytics & Risk Monitoring Portfolio

Project Overview:

The Bank Loan Analysis project was developed to evaluate lending performance, monitor loan portfolio health, and identify key business trends. Data was analyzed using SQL Server and visualized through Power BI dashboards. The project focuses on loan applications, funded amounts, repayments, interest rates, debt-to-income ratios, and borrower risk classification.

Business Problem:

Financial institutions require a centralized reporting solution to track lending activities, assess repayment performance, identify risky loans, and understand customer borrowing behavior. The challenge was to transform raw loan data into actionable insights for management and stakeholders

Project Workflow :

1. Imported loan dataset into Oracle SQL.
2. Created database and tables.
3. Performed data validation and cleaning.
4. Wrote SQL queries to calculate KPIs and business metrics.
5. Connected Power BI to Oracle SQL.
6. Built interactive dashboards and visualizations.
7. Validated dashboard results against SQL outputs.

Firing SQL Queries to Solve the Business Problems:

A. BANK LOAN REPORT | SUMMARY

KPI's:

Total Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM financial_loan;
```

Total_Applications
38576

MTD Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

Total_Applications
4314

PMTD Loan Applications

```
SELECT COUNT(id) AS Total_Applications
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

Total_Applications
4035

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount

FROM financial_loan;
```

Total_Funded_Amount
435757075

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount

FROM financial_loan

WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

Total_Funded_Amount
53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount

FROM financial_loan

WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

Total_Funded_Amount
47754825

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected

FROM financial_loan;
```

Total_Amount_Collected
473070933

MTD Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected
```

```
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

Total_Amount_Collected
58074380

PMTD Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

Total_Amount_Collected
50132030

Average Interest Rate

```
SELECT AVG(int_rate) * 100 AS Avg_Int_Rate
FROM financial_loan;
```

Avg_Int_Rate
12.0488314172048

MTD Average Interest

```
SELECT AVG(int_rate) * 100 AS MTD_Avg_Int_Rate
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

MTD_Avg_Int_Rate
12.3560408676042

PMTD Average Interest

```
SELECT AVG(int_rate) * 100 AS PMTD_Avg_Int_Rate
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

PMTD_Avg_Int_Rate
11.9417175498261

Avg DTI

```
SELECT AVG(dti) * 100 AS Avg_DTI
FROM financial_loan;
```

Avg_DTI
13.3274331211432

MTD Avg DTI

```
SELECT AVG(dti) * 100 AS MTD_Avg_DTI

FROM financial_loan

WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

MTD_Avg_DTI
13.6655377880425

PMTD Avg DTI

```
SELECT AVG(dti) * 100 AS PMTD_Avg_DTI

FROM financial_loan

WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

PMTD_Avg_DTI
13.3027335836364

GOOD LOAN ISSUED

Good Loan Percentage

```
SELECT
    (COUNT(CASE
              WHEN loan_status IN ('Fully Paid', 'Current')
              THEN id
           END) * 100.0) / COUNT(id) AS Good_Loan_Percentage
FROM financial_loan;
```

Good_Loan_Percentage
86.175342181667

Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications
FROM financial_loan
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_Applications
33243

Good Loan Funded Amount

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount
FROM financial_loan
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_Funded_amount
370224850

Good Loan Amount Received

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount
FROM financial_loan
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_amount_received
435786170

BAD LOAN ISSUED

Bad Loan Percentage

```
SELECT
    ROUND(
        SUM(CASE
                WHEN loan_status = 'Charged Off'
                THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
        2
    ) AS Bad_Loan_Percentage
FROM financial_loan;
```

Bad_Loan_Percentage
13.824657818332

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications

FROM financial_loan

WHERE loan_status = 'Charged Off';
```

Bad_Loan_Applications
5333

Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_Amount
FROM financial_loan
WHERE loan_status = 'Charged Off';
```

Bad_Loan_Funded_amount
65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_Amount_Received
FROM financial_loan
```

```
WHERE loan_status = 'Charged Off';
```

Bad_Loan_amount_received
37284763

LOAN STATUS

```
	SELECT
    loan_status,
    COUNT(id) AS LoanCount,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    ROUND(AVG(int_rate) * 100, 2) AS Interest_Rate,
    ROUND(AVG(dti) * 100, 2) AS DTI
FROM financial_loan
GROUP BY loan_status;
```

	loan_status	LoanCount	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

MTD LOAN STATUS REPORT (December)

```
SELECT
    loan_status,
    SUM(total_payment) AS MTD_Total_Amount_Received,
    SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM financial_loan
WHERE EXTRACT(MONTH FROM issue_date) = 12
GROUP BY loan_status;
```

loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
Fully Paid	47815851	41302025
Charged Off	5324211	8732775
Current	4934318	3946625

B. BANK LOAN REPORT | OVERVIEW

MONTH

```
SELECT
    EXTRACT(MONTH FROM issue_date) AS Month_Number,
    TO_CHAR(issue_date, 'Month') AS Month_Name,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY
    EXTRACT(MONTH FROM issue_date),
    TO_CHAR(issue_date, 'Month')
ORDER BY
    EXTRACT(MONTH FROM issue_date);
```

	Month_Munber	Month_name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

STATE

```
SELECT
    address_state AS State,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY address_state
```

```
ORDER BY address_state;
```

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660
24	MN	592	6302600	6750746
25	MO	660	7151175	7692732
26	MS	19	139125	149342
27	MT	79	829525	892047
28	NC	759	8787575	9534813
29	NE	5	31700	24542
30	NH	161	1917900	2101386
31	NJ	1822	21657475	23425159
32	NM	183	1916775	2084485
33	NV	482	5307375	5451443
34	NY	3701	42077050	46108181
35	OH	1188	12991375	14330148
36	OK	293	3365725	3712649
37	OR	436	4720150	4966903
38	PA	1482	15826525	17462908
39	RI	196	1883025	2001774

TERM

```
SELECT
    term AS Term,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY term
```

```
ORDER BY term;
```

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

EMPLOYEE LENGTH

```
SELECT
        emp_length AS Employee_Length,
        COUNT(id) AS Total_Loan_Applications,
        SUM(loan_amount) AS Total_Funded_Amount,
        SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length
```

Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
< 1 year	4575	44210625	47545011
1 year	3229	32883125	35498348
10+ years	8870	116115950	125871616
2 years	4382	44967975	49206961
3 years	4088	43937850	47551832
4 years	3428	37600375	40964850
5 years	3273	36973625	40397571
6 years	2228	25612650	27908658
7 years	1772	20811725	22584136
8 years	1476	17558950	19025777
9 years	1255	15084225	16516173

PURPOSE

```
SELECT
    purpose AS Purpose,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY purpose
ORDER BY purpose;
```

PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
car	1497	10223575	11324914
credit card	4998	58885175	65214084
Debt consolidation	18214	232459675	253801871
educational	315	2161650	2248380
home improvement	2876	33350775	36380930
house	366	4824925	5185538
major purchase	2110	17251600	18676927
medical	667	5533225	5851372
moving	559	3748125	3999899
other	3824	31155750	33289676
renewable_energy	94	845750	898931
small business	1776	24123100	23814817
vacation	352	1967950	2116738
wedding	928	9225800	10266856

HOME OWNERSHIP

```
SELECT
    home_ownership AS Home_Ownership,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY home_ownership
ORDER BY home_ownership;
```

Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
MORTGAGE	17198	219329150	238474438
NONE	3	16800	19053
OTHER	98	1044975	1025257
OWN	2838	29597675	31729129
RENT	18439	185768475	201823056

POWER BI DASHBOARD

DASHBOARD 1: SUMMARY

Developed an executive dashboard that provides a high-level overview of lending performance. It highlights loan application trends, funded amounts, repayment status, interest rates, DTI metrics, and loan quality indicators. Good loans were classified as Current or Fully Paid, while Charged-Off loans were classified as bad loans

Key Performance Indicators (KPIs) Requirements:

1. Total Loan Applications: We need to calculate the total number of loan applications received during a specified period. Additionally, it is essential to monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).

2. Total Funded Amount: Understanding the total amount of funds disbursed as loans is crucial. We also want to keep an eye on the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes in this metric.

3. Total Amount Received: Tracking the total amount received from borrowers is essential for assessing the bank's cash flow and loan repayment. We should analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.

4. Average Interest Rate: Calculating the average interest rate across all loans, MTD, and monitoring the Month-over-Month (MoM) variations in interest rates will provide insights into our lending portfolio's overall cost.

5. Average Debt-to-Income Ratio (DTI): Evaluating the average DTI for our borrowers helps us gauge their financial health. We need to compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.

Good Loan vs Bad Loan KPI's

Good Loan:

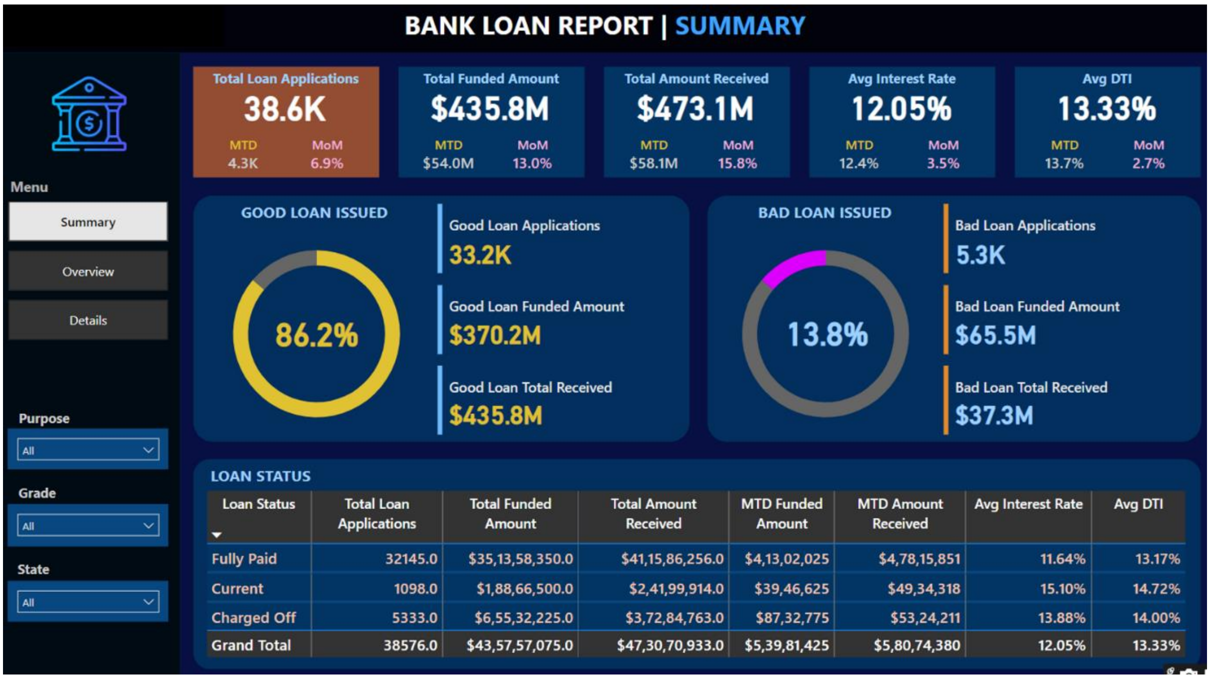
1. Good Loan Application Percentage
2. Good Loan Applications
3. Good Loan Funded Amount
4. Good Loan Total Received Amount

Bad Loan

1. Bad Loan Application Percentage
2. Bad Loan Applications
3. Bad Loan Funded Amount
4. Bad Loan Total Received Amount

Loan Status Grid View

In order to gain a comprehensive overview of our lending operations and monitor the performance of loans, we aim to create a grid view report categorized by 'Loan Status.' By providing insights into metrics such as 'Total Loan Applications,' 'Total Funded Amount,' 'Total Amount Received,' 'Month-to-Date (MTD) Funded Amount,' 'MTD Amount Received,' 'Average Interest Rate,' and 'Average Debt-to-Income Ratio (DTI),' this grid view will empower us to make data-driven decisions and assess the health of our loan portfolio.

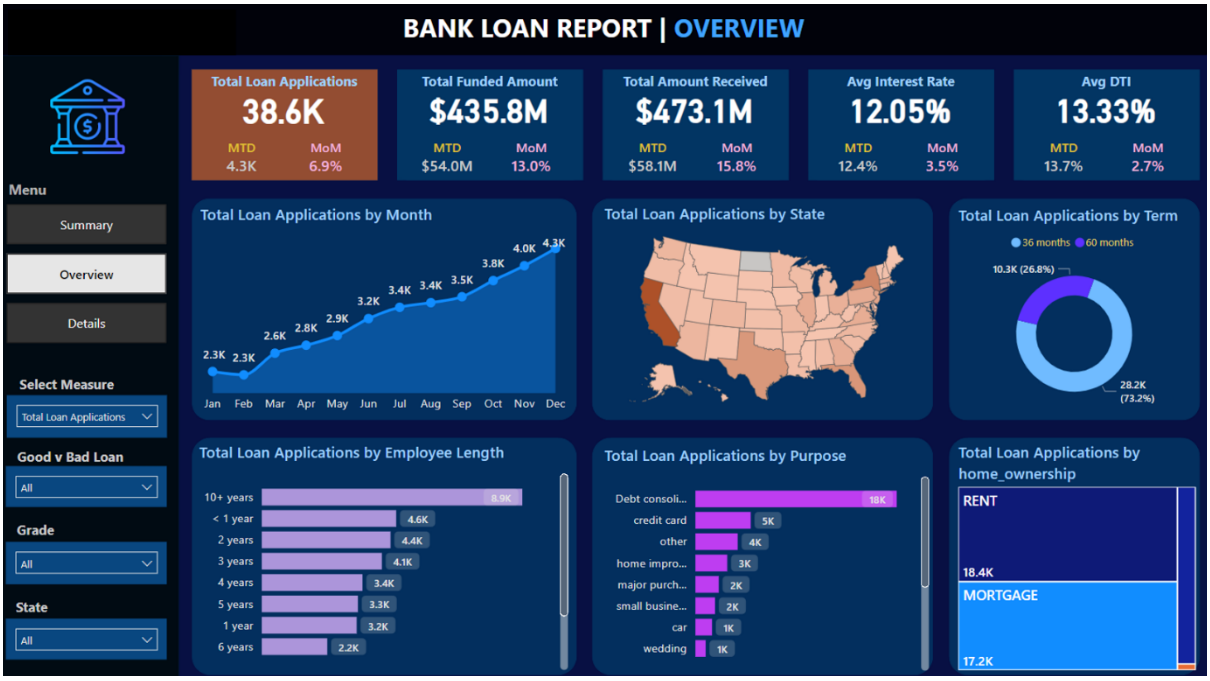


<u>DASHBOARD 2: OVERVIEW</u>

Interactive visualizations were created to identify business trends and borrower behavior. Analyses included monthly lending trends, state-wise loan distribution, loan term analysis, employment length impact, loan purpose breakdown, and home ownership analysis.

CHARTS

1. **Monthly Trends by Issue Date (Line Chart):** To identify seasonality and long-term trends in lending activities
2. **Regional Analysis by State (Filled Map):** To identify regions with significant lending activity and assess regional disparities
3. **Loan Term Analysis (Donut Chart):** To allow the client to understand the distribution of loans across various term lengths.
4. **Employee Length Analysis (Bar Chart):** How lending metrics are distributed among borrowers with different employment lengths, helping us assess the impact of employment history on loan applications.
5. **Loan Purpose Breakdown (Bar Chart):** Will provide a visual breakdown of loan metrics based on the stated purposes of loans, aiding in the understanding of the primary reasons borrowers seek financing.
6. **Home Ownership Analysis (Tree Map):** For a hierarchical view of how home ownership impacts loan applications and disbursements.
Metrics to be shown: *'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'*



DASHBOARD 3: DETAILS

A detailed reporting dashboard was designed to provide a complete view of loan-level information. Users can explore detailed borrower and loan performance metrics to support operational and strategic decision-making.

GRID

Need for a comprehensive 'Details Dashboard' that provides a consolidated view of all the essential information within our loan data. This Details Dashboard aims to offer a holistic snapshot of key loan-related metrics and data points, enabling users to access critical information efficiently.

Objective:

The primary objective of the Details Dashboard is to provide a comprehensive and user-friendly interface for accessing vital loan data. It will serve as a one-stop solution for users seeking detailed insights into our loan portfolio, borrower profiles, and loan *performance.*

id	purpose	home_ownership	grade	sub_grade	issue_date	Funded Amount	Int Rate	Sum of installment	Amount Collection
54734	Debt consolidation	RENT	B	B4	09 August 2021	$25,000.0	11.89%	$829.1	$29,330.0
55742	credit card	RENT	B	B5	08 May 2021	$7,000.0	10.71%	$228.2	$8,216.0
57245	Debt consolidation	OWN	C	C2	10 March 2021	$1,200.0	13.11%	$40.5	$1,458.0
57416	Debt consolidation	RENT	C	C3	09 November 2021	$10,800.0	13.57%	$366.9	$13,208.0
58915	Debt consolidation	RENT	B	B3	08 April 2021	$7,500.0	10.08%	$162.3	$5,844.0
59006	credit card	MORTGAGE	C	C5	09 September 2021	$3,000.0	14.26%	$102.9	$3,705.0
61390	credit card	MORTGAGE	A	A5	10 February 2021	$4,000.0	7.88%	$125.1	$4,452.0
61419	Debt consolidation	RENT	D	D2	10 February 2021	$5,600.0	14.96%	$194.0	$6,475.0
62102	Debt consolidation	RENT	B	B1	10 April 2021	$3,200.0	9.88%	$103.1	$3,414.0
65426	car	MORTGAGE	B	B1	09 August 2021	$4,000.0	11.14%	$131.2	$2,755.0
65640	home improvement	MORTGAGE	C	C2	08 May 2021	$5,000.0	11.34%	$87.2	$3,154.0
66431	Debt consolidation	RENT	B	B5	09 February 2021	$2,525.0	12.21%	$84.1	$3,028.0
66749	Debt consolidation	MORTGAGE	C	C4	08 December 2021	$10,625.0	13.47%	$360.4	$12,975.0
66943	Debt consolidation	RENT	B	B4	10 August 2021	$2,800.0	11.49%	$61.6	$3,144.0
66964	Debt consolidation	MORTGAGE	D	D3	08 June 2021	$7,500.0	13.24%	$253.6	$9,129.0
67503	Debt consolidation	MORTGAGE	A	A4	09 October 2021	$10,000.0	8.59%	$316.1	$11,280.0
68163	small business	MORTGAGE	A	A3	10 February 2021	$3,000.0	7.14%	$92.8	$3,342.0
68381	Debt consolidation	RENT	A	A5	08 March 2021	$6,625.0	8.63%	$209.5	$7,542.0

Key Insights Generated:

- Identified overall loan portfolio growth trends.
- Measured borrower repayment performance.
- Evaluated good versus bad loan distribution.
- Determined regional lending patterns across states.
- Analyzed borrower employment history and loan purposes.
- Assessed financial risk through DTI and interest rate analysis.

Business Impact & Solution: The solution provides management with a centralized view of loan performance, helping identify high-risk loans, monitor repayment efficiency, optimize lending strategies, and improve portfolio management. The dashboards enable faster and more informed decision-making through real-time analytics

Project Summary: This project demonstrates end-to-end Data Analytics capabilities, including data extraction, SQL analysis, KPI development, dashboard design, and business storytelling. By combining Oracle SQL and Power BI, the project converts raw banking data into actionable insights that improve operational visibility and support strategic decisions